

WILLIAMSBURG, SC, USA

WMO: 720604

Lat: 33.717N

Lon: 79.850W

Elev: 20

StdP: 101.08

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00196

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.8	(c) -2.3	(d) -12.5	(e) 1.3	(f) 0.6	(g) -10.0	(h) 1.6	(i) 2.8	(j) 8.3	(k) 11.5	(l) 7.5	(m) 11.9	(n) 0.5	(o) 290	(p) 0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	35.3	25.2	33.9	24.8	32.7	24.6	28.4	32.7	27.4	30.3	26.6	29.5	2.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.4	30.2	26.4	21.9	28.6	26.0	21.4	28.2	91.3	31.7	86.4	29.9	82.8	30.0	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.0	(b) 5.6	(c) 4.8	(e) -7.8	(f) 36.7	(g) 2.5	(h) 1.9	(i) -9.6	(j) 38.1	(k) -11.1	(l) 39.2	(m) -12.5	(n) 40.3	(o) -14.4	(p) 41.7		
(4) 7.0	(5) 5.6	(6) 4.8	(7) -7.8	(8) 36.7	(9) 2.5	(10) 1.9	(11) -9.6	(12) 38.1	(13) -11.1	(14) 39.2	(15) -12.5	(16) 40.3	(17) -14.4	(18) 41.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	8.0	10.4	12.9	17.8	22.5	26.4	27.3	26.4	24.0	18.5	12.2	10.0
	DBStd	7.97	5.67	5.04	5.14	4.08	3.19	2.27	1.93	1.74	2.55	4.19	4.47	5.25
	HDD10.0	282	108	52	29	1	0	0	0	0	0	1	24	67
	HDD18.3	1299	324	225	180	58	7	0	0	0	1	50	191	263
	CDD10.0	3233	44	64	119	237	389	492	536	509	421	266	89	67
	CDD18.3	1208	2	4	12	43	138	242	278	251	172	56	6	5
	CDH23.3	11314	13	32	125	456	1253	2468	2817	2254	1387	452	41	17
	CDH26.7	4697	0	1	19	112	478	1152	1319	964	541	108	2	1
Wind	WSAvg	1.5	1.7	2.0	2.1	2.1	1.8	1.4	1.4	1.1	1.2	1.2	1.3	1.4
Precipitation	PrecAvg	1283	112	95	113	80	94	135	131	158	110	86	66	90
	PrecMax	1836	210	245	233	236	289	358	313	326	343	320	180	254
	PrecMin	923	32	26	22	8	20	38	11	46	1	0	14	17
	PrecStd	196	48	48	54	50	50	74	69	69	72	81	40	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	26.4	28.8	31.2	35.0	37.4	37.4	35.2	33.8	31.4	27.1	25.2
		MCWB	19.7	18.4	19.6	21.6	22.1	27.8	27.7	25.8	23.6	23.2	20.8	20.0
	2%	DB	22.1	23.9	27.1	29.9	32.6	35.2	35.9	33.7	32.5	29.6	24.7	22.7
		MCWB	18.1	17.8	18.5	20.2	21.7	24.9	26.1	25.2	23.8	22.2	19.3	19.4
	5%	DB	19.9	22.2	24.8	27.7	31.3	33.0	33.9	32.6	31.3	27.5	22.4	21.2
		MCWB	16.0	16.3	16.9	19.2	21.8	24.5	25.6	25.3	23.4	21.3	18.2	18.7
	10%	DB	17.4	19.9	22.3	26.2	29.8	32.4	32.5	32.0	30.1	26.1	21.0	19.0
		MCWB	14.5	15.0	16.2	18.4	21.8	24.1	25.5	25.3	23.3	20.8	17.5	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	20.4	22.3	24.2	26.2	29.7	30.3	28.4	27.5	25.8	22.9	21.5
		MCDB	22.9	23.5	25.5	28.2	29.8	34.9	34.3	32.5	30.7	27.0	24.3	23.4
	2%	WB	19.1	19.0	20.5	22.8	24.5	28.3	28.7	27.5	26.4	24.6	21.2	20.6
		MCDB	21.4	22.7	23.7	26.1	28.3	32.9	33.0	30.6	28.6	27.1	22.4	22.3
	5%	WB	17.3	17.5	18.8	21.8	23.5	26.8	27.4	26.9	25.6	23.3	19.8	19.0
		MCDB	18.3	20.4	22.6	25.4	27.5	29.7	30.2	29.8	28.1	26.3	21.4	20.8
	10%	WB	15.0	16.0	17.4	20.5	22.8	25.5	26.6	26.2	24.6	22.3	17.9	17.2
		MCDB	17.3	18.7	20.5	23.4	26.6	29.0	29.8	28.9	28.0	24.9	19.9	18.4
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.4	13.0	13.6	11.8	11.2	10.7	10.1	11.0	12.6	13.0	11.1
		MCDBR	13.9	14.6	15.2	15.2	14.1	12.4	12.4	11.5	12.8	13.5	14.3	12.7
		MCWBR	9.2	9.0	7.7	6.9	5.1	4.4	4.7	4.8	5.5	7.1	9.2	9.2
	5% WB	MCDBR	11.2	12.8	12.6	11.5	10.7	11.8	10.5	9.6	10.2	10.2	10.7	11.1
		MCWBR	8.8	9.1	7.5	6.5	4.9	5.5	5.2	5.0	5.0	6.3	8.5	8.8
Clear-Sky Solar Irradiance	taub	0.322	0.329	0.364	0.406	0.473	0.499	0.546	0.517	0.433	0.393	0.345	0.320	
	taud	2.511	2.484	2.414	2.283	2.145	2.120	2.007	2.079	2.307	2.388	2.482	2.550	
	Ebn at Noon	893	922	911	882	823	799	759	774	830	838	856	875	
	Edh at Noon	84	96	111	133	154	158	175	161	123	105	86	77	
All-Sky Solar Radiation	RadAvg	2.79	3.42	4.54	5.76	6.21	6.40	6.13	5.52	4.75	4.00	3.08	2.47	
	RadStd	0.15	0.35	0.37	0.39	0.47	0.45	0.38	0.28	0.40	0.47	0.24	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+183	+131

Nomenclature: See separate page